

Offline Preparation

Dataset Pipeline

CICIoT2023
(105 IoT devices,
33 attack types)
Kill-chain projection
($\times 5$ stages)
Splits: train / val /
test / ood

Reactive Escalation Attacker

Proximity-coupled
kill-chain process
Strictly sequential;
defender-driven
de-escalation;
non-terminal IMPACT;
final mitigation turn

Stage Detector

RandomForest (tuned)
Input: 29-dim
per-flow features
Output: $\hat{s} \in$
 $\{\text{benign}, \dots, \text{impact}\}$

eval-time

Realization Engine

Session-coherent,
aliasing-aware
sampling of 29-dim
feature vectors

features

AdversarialIoTEnv

Hidden attack stage s_t
Kill-chain POMDP
(S, A, O, P, Z, R, γ)

Outcome Reward

Action cost
+ impact accounting
+ prevention bonus

o_t

a_t

Blue-Team Agent (Defender)

DQN / PPO / A2C
(Stable Baselines3
MlpPolicy)
10 seeds $\times$ 5M steps (no early stopping)

r_t

$o_t = \phi(x_{t-w+1:t}) \in \mathbb{R}^{290}$
(29 features + Δ , $\times 5$ -row window)

Online RL Training Loop

Benchmark & Ablation

Trained DRL (DQN / PPO / A2C) vs. Random — Always-Observe — Always-Block — RF-Acting — Oracle
Reward-coupling ablation — Environment-difficulty sweep — OOD injection evaluation